3 Points Each

1998

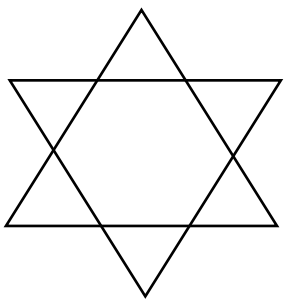
1. At the zoo, Jacob saw kangaroos for the first time. He noticed that each kangaroo had four legs, two ears, and one tail. For fun, he counted all the legs, ears, and tails, and got the number 63. How many kangaroos did he see?

a) 6
b) 7
c) 9
d) 10
e) 12

2. David wanted to place parentheses in the expression 6 x 8 + 20 ÷ 4 – 2 in such a way as to get 58 as the result. Which of the ways shown below would give him this result?

a) 6 x (8 + 20) ÷ 4 – 2
b) (6 x 8 + 20 ÷ 4) – 2
c) (6 x 8 + 20) ÷ 4 – 2
d) 6 x 8 + 20 ÷ (4 – 2)
e) 6 x (8 + 20 ÷ 4) – 2

3. How many triangles can you see in the picture?



a) 2
b) 6
c) 8
d) 10
e) 12

4. Faith lives in a tall building in apartment number 17. There are stores on the first floor. Above the stores, there are 3 apartments on each floor, numbered consecutively. On what floor does Faith live?

a) 4th
b) 5th
c) 6th
d) 7th
e) 8th

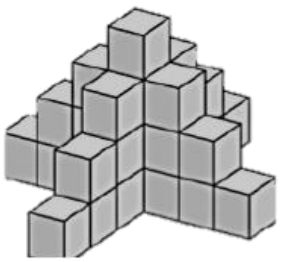
5. What number do we need to place in the blank to make 12 x 12 x 12 = 6 x ___ x 6 true?

a) 12
b) 24
c) 48
d) 72
e) 60

6. How many three-digit numbers can you create using the digits 3, 0, and 7, and using each digit only once?

a) 2
b) 3
c) 4
d) 5
e) 6

7. Out of how many blocks is this tower built?



a) 20
b) 22
c) 25
d) 28
e) 30

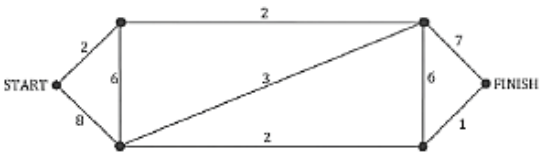
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8. I chose a number, then I multiplied it by 3 and added 18 to the result. I got 180. What was the number I chose?

a) 52
b) 54
c) 50
d) 48
e) 62

4 Points Each

9. A kangaroo is traveling from the START to FINISH using the paths shown in the picture. Each segment is marked with the time (in minutes) which the kangaroo needs to travel that segment. What is the shortest time needed for the kangaroo to reach FINISH?



a) 11 minutes
b) 8 minutes
c) 10 minutes
d) 18 minutes
e) 6 minutes

10. Joanna baked some cookies. She tried to divide them evenly, first between 2 plates, then between 3 plates, and finally between 4 plates. Each time she had one cookie left over. What is the smallest number of cookies Joanna could have baked?

a) 9
b) 10
c) 11
d) 12
e) 13

11. I chose a certain number. I then subtracted 40 from it. Then added 2000 and as a result I now have 3250. What number did I choose in the beginning?

a) 2040
b) 1960
c) 1290
d) 3210
e) 1250

12. On Monday morning, a snail fell down a well which is 5 meters deep. During the day, it climbs up 2 meters, but during the night it slides down 1 meter. On what day of the week will the snail get out of the well?

a) Tuesday
b) Wednesday
c) Thursday
d) Friday
e) Monday

13. Adam cut five identical square sheets of paper into two pieces. From which of the five pieces below was the piece marked with Z cut?



a) 
b) 
c) 
d) 
e) 

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5 Points Each

14. There were 31 runners competing in s race. The number of runners who finished before John is four times smaller than the number of runners who finished later than John. In what place did John finish?

a) 6
b) 7
c) 8
d) 20
e) 21

15. Half a loaf of bread costs 6 pence more than one-fourth of a loaf bread. How many pence does a whole loaf of bread cost? (Note: A pence is an English coin)

a) 6
b) 12
c) 18
d) 24
e) 30

16. Little Kangaroo's clock fell from the wall. The Kangaroo does not know which side should go up. Can you help her and tell what time is?

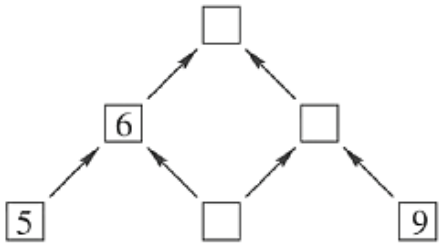


a) 11:45
b) 9:00
c) 8:25
d) 6:35
e) 3:00

17. Write the numbers into the blank boxes in the pyramid according to the pattern shown below.



What number is at the top of the pyramid?



a) 5
b) 7
c) 8
d) 9
e) 12

18. There are 15 balls in a box: white balls, red balls, and black balls. The number of white balls is 7 times greater than the number of red balls. How many black balls are there in the box?

a) 1
b) 3
c) 5
d) 7
e) 9

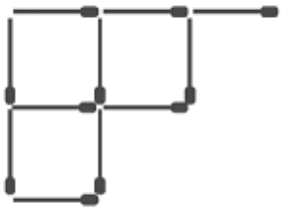
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19. Paul was going to buy 4 servings of ice cream, but he was 80 cents short. So, he bought 3 servings and had 30 cents left. What was the price of one serving of ice cream?

a) 70 cents
b) 80 cents
c) 90 cents
d) 1 dollar
e) 1 dollar and 10 cents

20. We are making a square "cheeseboard" using matches that are 5 centimeters long. One side of the cheeseboard will be 1 meter long.

The picture shows the upper left-hand corner of the board.



How many matches will we use? (Note: 1 m = 100 cm)

a) 400
b) 480
c) 640
d) 840
e) 960

21. How many three-digit numbers are there that have the sum of their digits equal to 5?

(For example , 122 is such a number , because 1 + 2 + 2 = 5)

a) 10
b) 15
c) 20
d) 25
e) 30

22. Ann is 3 years older than Baro and 2 years younger than Cali. Dorothy is 1 year younger than Barb. How much older is Cali than Dorothy?

a) 5 years
b) 6 years
c) 4 years
d) 2 years
e) They are the same age.

23. In a certain soccer tournament, the winning team gets 3 points, the losing team gets 0 points, and in the case of a tie both teams get 1 point each. My team played 31 games and received 64 points. 7 of the games were ties. How many games did my team lose?

a) 0
b) 5
c) 19
d) 21
e) 24

24. The number 123456789101112131415 16... is made by writing consecutive natural numbers in order starting from 1. Which natural number does the digit in the 180th place belong to?

a) 92
b) 93
c) 94
d) 95
e) 96

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Answers key:

1. C
2. D
3. C
4. D
5. C
6. C
7. D
8. B
9. C
10. E
11. C
12. C
13. A
14. B
15. D
16. E
17. B
18. D
19. E
20. D
21. B
22. B
23. B
24. D

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3 Point Solutions

1 **(C) 9**
Each kangaroo had four legs, two ears, and one tail, so Bob counted 4 + 2 + 1 = 7 parts for every kangaroo. 63 ÷ 7 = 9, so Bob saw 9 kangaroos. Check:



7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 = 9 × 7 = 63

2 **(D) 6 × 8 + 20 ÷ (4 – 2)**
Remember the order of operations. First do what is in the parentheses. Then, from left to right, do multiplication and division, and finally do addition and subtraction, also from left to right.

(A) 6 × (8 + 20) ÷ 4 – 2 = 6 × 28 ÷ 4 – 2 = 168 ÷ 4 – 2 = 42 – 2 = 40

(B) (6 × 8 + 20 ÷ 4) – 2 = (48 + 20 ÷ 4) – 2 = (48 + 5) – 2 = 53 – 2 = 51

(C) (6 × 8 + 20) ÷ 4 – 2 = (48 + 20) ÷ 4 – 2 = 68 ÷ 4 – 2 = 17 – 2 = 15

(D) 6 × 8 + 20 ÷ (4 – 2) = 6 × 8 + 20 ÷ 2 = 48 + 10 = 58

(E) 6 × (8 + 20 ÷ 4) – 2 = 6 × (8 + 5) – 2 = 6 × 13 – 2 = 78 – 2 = 76

3 **(C) 8**
There are two large overlapping triangles (one marked red and one marked purple in the picture) and six small triangles (marked green).



4 **(D) 7th**
Apartments 1, 2, 3 are on the 2nd floor; apartments 4, 5, 6 are on the 3rd floor; apartments 7, 8, 9 are on the 4th floor; apartments 10, 11, 12 are on the 5th floor; apartments 13, 14, 15 are on the 6th floor; and apartments 16, 17, 18 are on the 7th floor. Mary lives on the 7th floor.

5 **(C) 48**
12 × 12 × 12 = (6 × 2) × 12 × (2 × 6) = 6 × (2 × 12 × 2) × 6 = 6 × 48 × 6

6 **(C) 4**
A three-digit number cannot start with a zero. So, the only possible numbers are 307, 370, 703, and 730.

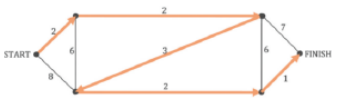
7 **(D) 28**
In the very center of the tower there are 4 vertical blocks, and around it there are 4 staircase-like structures, each made of 6 blocks. The total number of blocks is 4 + 4 × 6 = 28.



8 **(B) 54**
Work backwards. The last step was to add 18, so subtract 18 from the final number. 180 – 18 = 162. Before that, the number was multiplied by 3, so divide by 3. 162 ÷ 3 = 54. 54 was the number that was chosen at the beginning.

4 Point Solutions

9 **(C) 10 minutes**
Try different paths. Zigzagging through the middle part of the figure gives the shortest time.



10 **(E) 13**
Possible numbers of cookies for 3 plates when 1 cookie is always left over are: 4, 7, 10, 13, 16, 19, 22, 25 , 28, ... (multiples of 3 plus 1).

Possible numbers of cookies for 4 plates when 1 cookie is always left over are: 5, 9, 13, 17, 21, 25, 29, ... (multiples of 4 plus 1).

The smallest number on both lists is 13. If Joanna divides evenly 13 cookies between 2 plates, 1 cookie is also left over.

11 **(C) 1290**
To find a solution, reverse your calculations. If something was added, subtract it; if it was subtracted, add it back. 3250 – 2000 = 1250 and 1250 + 40 = 1290.

12 **(C) Thursday**
The well is 5 meters deep. On Monday the snail goes up 2 meters but slides 1 meter down. It starts Tuesday with still 4 meters to go up, moves 2 meters up and slides 1 meter down, so on Wednesday it starts with 3 meters left to climb. Again it goes up 2 meters and slides 1 meter down. On Thursday morning it is 2 meters from the top which it climbs during the day and finally gets out of the well.

13 **(A)**



Rotate piece Z so it is upside-down, and then piece Z fits with piece (A).



14 **(B) 7**
There were 30 runners if we don't count John. They can be divided into 5 groups of equal size: 1 group that finished before him and 4 groups that finished after him. There were 6 runners in each group since 5 × 6 = 30. There was one group of 6 runners before John, so he finished in the 7th place.

15 **(D) 24**
The difference between half and one-fourth of a loaf of bread is one-fourth of a loaf of bread. Since one fourth of a loaf costs 6 pence, the whole loaf costs 4 × 6 pence = 24 pence.

16 **(E) 3:00**
The hour hand (the thicker, shorter hand) is pointing directly at an hour mark. This only happens when the time is "on the hour." So, the minute hand is pointing at 12. Rotate the clock so that 12 is on top, and you can see that the hour hand is pointing to the third hour mark, and the time is 3:00.



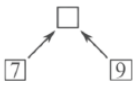
5 Point Solutions

17 **(B) 7**
First, look at this part of the pyramid:



The pattern shows us to add the two lower numbers together and then divide by 2 to get the top number. In this blank box, we have to write a number such that 5 plus this number is the double of 6. 12 ÷ 2 = 6, and 5 + 7 = 12, so 7 is that number.

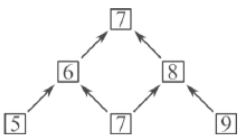
Now that we know that there is a 7 in the middle bottom box, let's look at this part:



In the blank box we have to write 8, since $\frac{(7+9)}{2} = 8$

At the top of the pyramid we have to write $\frac{(6+8)}{2}$ which is 7.

Here is the full pyramid:

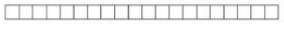


18 **(D) 7**
If there were 2 or more red balls, then the number of white balls would be at least 7 × 2 = 14, and the number of red and white balls would already be more than 15. Thus, there is only 1 red ball and 7 × 1 = 7 white balls. 1 + 7 + 7 = 15, so the other 7 balls are black.

19 **(E) 1 dollar and 10 cents**
Notice that if Paul added 80 cents to the 30 cents he had left, he would have enough money for another serving of ice cream. So, one serving costs 80 + 30 = 110 cents, which is equal to 1 dollar and 10 cents.

20 **(D) 840**
1 meter = 100 cm and 20 × 5 cm = 100 cm, so the size of the chessboard is 20 × 20 matches.

We will count vertical matches in each row and horizontal matches in each column.



If you have a row of 20 adjacent squares, then the number of vertical matches is 20 + 1 = 21.

There are 20 rows, so the number of all vertical matches is 20 × 21.

We can count the number of horizontal matches in the same way, so 20 × 21 is the number of all horizontal matches. The number of all the matches is 2 × 20 × 21 = 840.

21 **(B) 15**
The numbers, in their natural order, are: 104, 113, 122, 131, 140; 203, 212, 221, 230; 302, 311, 320; 401, 410, and 500. There are 15 numbers on the list.

22 **(B) 6 years**
Look at the line below showing the age differences from youngest to oldest. Each girl is represented by her initial.



1 + 3 + 2 = 6, so Cali is 6 years older than Dorothy.

23 **(B) 5**
My team received 64 points for ties and wins. The team earned 7 points for 7 ties and 57 points for a number of wins (7 + 57 = 64). The team won 19 games since 19 × 3 = 57 and 7 games were ties, so my team lost 31 – (7 + 19) = 31 – 26 = 5 games.

24 **(D) 95**
The one-digit numbers (1 to 9) take up positions from 1 to 9. Then the two-digit numbers start in position 10, and each takes up 2 positions.

123456789101112131415 16...

There are 90 two-digit natural numbers (10 to 99 inclusive), which is 180 digits, which would make the last 9 of 99 be in the 189th positions.

...909192939495969 7899

Going back 9 positions to 180 we are at the 9 of 95. So, the natural number we are looking for is 95.

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